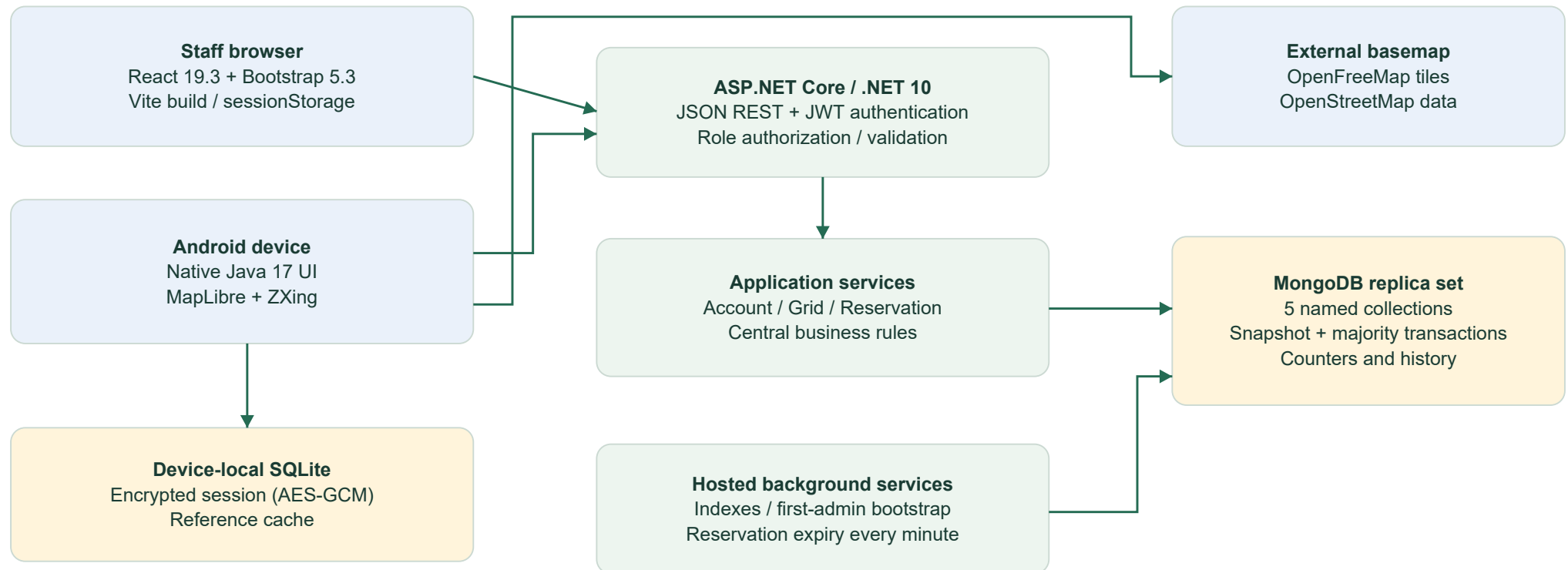


# System architecture

02 / Deployment containers, communication paths and the technology stack



Client-to-API: JSON REST; HTTPS for release. Database access is server-only. Maps are fetched directly by Android.

Solara is a client-server application with one central API and one authoritative MongoDB database. The API contains modular services, not separately deployed microservices.

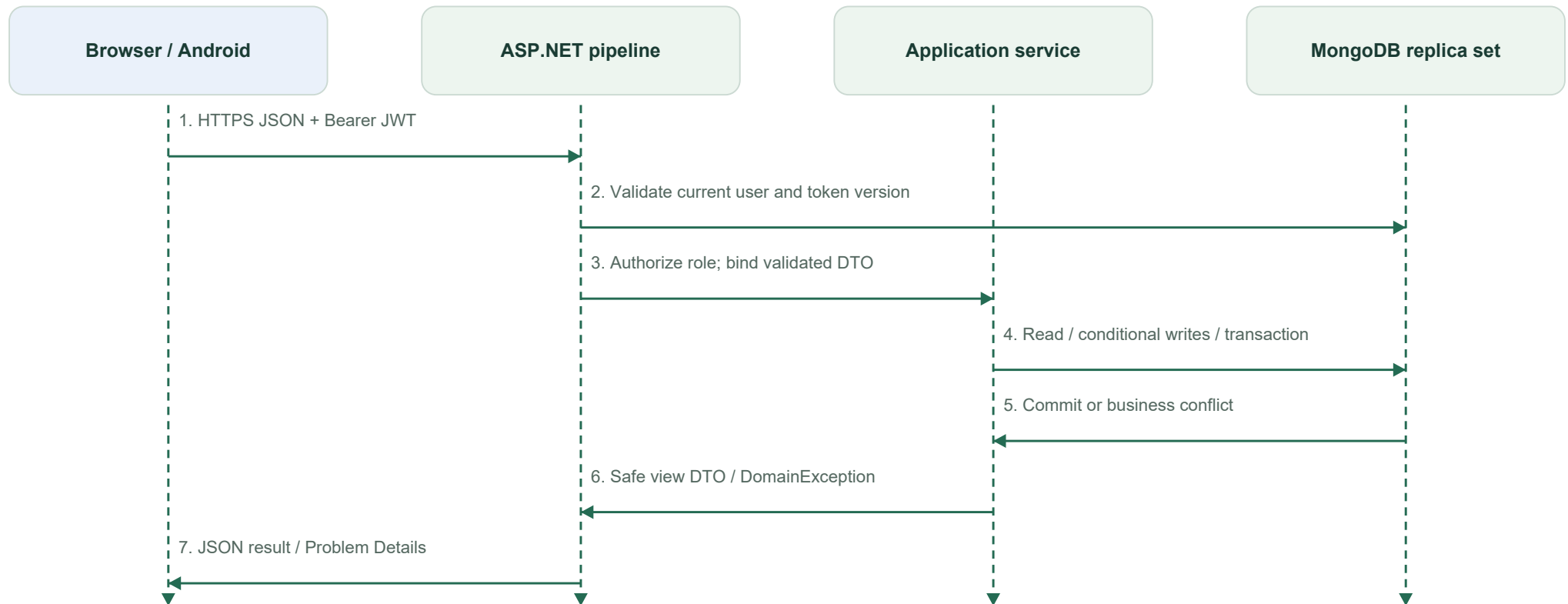
# Technology stack

Versions below are declared by the project manifests, not a claim about latest upstream releases.

| Layer                 | Technology / project version                                                                           | Purpose                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Web interface         | React + React DOM 19.3.0; Bootstrap 5.3.8; lucide-react ^0.468.0                                       | Staff dashboards, operational forms, account administration and payload verification.        |
| Web build             | Vite ^8.0.0; Node.js 22.12+ prerequisite; npm                                                          | Development server and production static bundle. API base URL supplied through VITE_API_URL. |
| Android               | Native Java 17; minSdk 26; compileSdk/targetSdk 36; workspace Gradle 8.13                              | Activities and native widgets. No React Native or Flutter dependency.                        |
| Map / QR              | MapLibre Android OpenGL 13.5.2; OpenFreeMap / OSM; ZXing embedded 4.3.0 + core 3.5.3                   | Map renderer, public basemap, QR rendering and camera scan.                                  |
| API                   | C#; ASP.NET Core / .NET 10; JWT Bearer package 10.0.12                                                 | REST controllers, dependency injection, validation, authorization, hosted workers.           |
| Authoritative data    | MongoDB 8.0+ replica set prerequisite; MongoDB.Driver 3.12.0                                           | Transactions across reservations, capacity counters and transactional audit events.          |
| Local data / security | Android SQLiteOpenHelper; Android Keystore AES-GCM; browser sessionStorage                             | Encrypted Android session, reference cache, browser-tab session state.                       |
| Hosting / checks      | Windows IIS + .NET 10 Hosting Bundle; Kestrel for local dev; .NET test runner; Vite build; Gradle lint | Deployment and verification. Exact installed tool versions may differ from manifest ranges.  |

# Runtime request architecture

Thin controllers delegate decisions to services; MongoDB remains unreachable from the client.



Pipeline order: error wrapper and response headers; CORS; authentication; authorization; rate limiter; mapped controllers. DatabaseInitializer and ReservationExpiryService run inside the API host.

# Development and IIS deployment

These are configured deployment paths; this document does not assert that the services are live at reading time.

| Component               | Development                                                          | IIS / release path                                                                                                          |
|-------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Web portal              | Vite at http://127.0.0.1:5173                                        | IIS static bundle; local assessment web binding http://127.0.0.1:8081.                                                      |
| REST API                | dotnet run / Kestrel; http://localhost:5080/api                      | ASP.NET Core behind IIS; local assessment API binding http://127.0.0.1:8080/api.                                            |
| Database                | Workspace Mongo at 127.0.0.1:27018, replicaSet=rs0                   | Configured Mongo connection string; production needs authenticated, restricted access and resilient replica-set deployment. |
| Android emulator        | http://10.0.2.2:5080/api for local dev                               | App default debug endpoint is http://10.0.2.2:8080/api for local IIS; adjustable in Connection settings.                    |
| Physical Android device | USB adb reverse can map device localhost:5080 to the development API | Use a reachable trusted HTTPS endpoint for release. Debug alone permits HTTP.                                               |
| Release configuration   | Ignored appsettings.Local.json and local tools                       | JWT/QR secrets and Mongo settings via secured host configuration; VITE_API_URL is baked into the web build.                 |

The publish script prepares artifacts; it is not automatic production deployment. IIS sites use separate application pools. A local single-member replica set enables transactions but does not provide multi-server availability.

# Trust boundaries and system scope

Implemented controls and explicit boundaries

| Boundary             | Implemented behavior / limit                                                                                                                                                                    |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Client to API        | JWT signature, issuer, audience and expiry validation; account Active status and TokenVersion checked for every authenticated request. Staff roles and prosumer ownership enforced server-side. |
| Credentials and QR   | ASP.NET password hashing. JWT and QR use independent keys of at least 32 bytes. QR uses HMAC-SHA256 and a current booking nonce; QR verification requires live database state.                  |
| API to database      | Pooled singleton MongoClient; snapshot read concern and majority write concern for transactions. Clients never receive Mongo connection credentials.                                            |
| Device local storage | Session JSON encrypted with a non-exportable Keystore key. Reference cache is not uniformly encrypted; cached data cannot authorize mutations or transfers.                                     |
| Maps / hardware      | Basemap calls go to OpenFreeMap. Registered pins come from the API. No real-time energy meter, charger, payment gateway or notification provider is integrated.                                 |

## Implementation references

These are repository paths, not external product documentation. The code is the source of truth for this guide.

backend/SolarTrading.Api/Program.cs and SolarTrading.Api.csproj

backend/SolarTrading.Api/Repositories/MongoStore.cs

frontend/web/package.json and src/api.js

frontend/android/app/build.gradle and ApiClient.java / LocalStore.java

scripts/start-local.ps1, scripts/publish.ps1 and docs/deployment/iis.md